

# Rohit Kumar Yadav

Uttarakhand (Dehradun)

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## EDUCATION

GRAPHIC ERA UNIVERSITY Dehradun, India

2023 – 2025

Master of Computer Applications | CGPA: 7.4

GRAPHIC ERA UNIVERSITY Dehradun, India

2020 – 2023

Bachelor of Science in Information Technology | CGPA: 7.6

## SKILLS SUMMARY

- **Programming & Databases:** Python, SQL, PostgreSQL, SQLAlchemy (Async)
- **GenAI & LLMs:** LangGraph, LangChain, LangSmith, AWS Bedrock, Advanced RAG, Tool Calling, MCP, LLM Agents
- **ML / DL:** Scikit-Learn, TensorFlow, PyTorch, Transformers, Hugging Face, LLaMA, LLM Fine-Tuning
- **NLP & Retrieval:** NLP Pipelines, Embeddings, Semantic Search, ChromaDB, FAISS, Vector Databases
- **Data & Visualization:** Pandas, NumPy, Matplotlib, Power BI
- **MLOPs & Infrastructure:** Docker, Kubernetes, MLflow, AWS (EC2, S3), Redis, GitHub Actions (CI/CD), Render
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code, PyCharm, Grafana
- **Backend & APIs:** FastAPI, Pydantic, REST APIs, JWT Authentication

## PROJECTS

### Agentic AI Assistant with MCP Integration | [GitHub](#) | [Demo](#)

**Tech Stack:** Python, LangGraph, LangSmith, FastAPI, MCP, Groq, Ollama, PostgreSQL, Docker, APIs

- Built an Agentic AI Assistant using LangGraph and MCP, integrating 7+ tools for expense management, web search, stock analysis, weather, and local events.
- Developed custom MCP servers with SQLite and remote MCP integrations, enabling autonomous tool execution through natural language.
- Integrated LangSmith, Groq, and Ollama for agent observability, workflow tracing, and improved debugging efficiency by ~60%.

### Multimodal AI System | [Live](#) | [GitHub](#) | [Demo](#)

**Tech Stack:** Python, LangChain, Streamlit, Chroma DB, Google Gemini (LLM), LangGraph, Embeddings

- Built and deployed a Multimodal RAG platform for querying YouTube videos, PDFs, and documents, indexing 1,000+ chunks for semantic retrieval.
- Implemented a **Corrective RAG (CRAG) pipeline** with embeddings, vector search, and retrieval validation, improving response accuracy by ~30%.
- Developed a LangGraph-based query router that dynamically selects knowledge sources, improving answer relevance by ~25%.

### Agentic Trip Planning System | [Live](#) | [GitHub](#) | [Demo](#)

**Tech Stack:** Python, LangGraph, FastAPI, Streamlit, Groq, LangSmith, SQLite, Geoapify, Tavily, Render

- Developed an AI-powered travel planning system using LangGraph and Groq LLMs to generate personalized multi-day itineraries based on user preferences, budget, interests, and travel dates.
- Integrated travel APIs for weather, hotels, attractions, geolocation, and transportation recommendations.
- Deployed on Render with persistent trip history, Markdown export, and LangSmith monitoring.

## EXPERIENCE

Data Analytics Intern | [Infyntrek Systèmes](#) | [Certificate](#)

Sep 2025-Dec 2025

- Performed data preprocessing, feature engineering, and EDA using Python and SQL for ML workflows.
- Built and evaluated ML/DL models using Scikit-Learn and TensorFlow, delivering actionable insights.

## HACKATHONS

- Participated in **HackerRank** Orchestrate June 2026, building and presenting an AI agent in a 24-hour hackathon.
- Built **RecruitRankAI** for the **INDIA.RUNS** AI Challenge by **Redrob**, leveraging Sentence Transformers, FAISS, and hybrid ranking to generate explainable Top-100 candidate recommendations under CPU-only, offline, and no-LLM competition constraints.

## CERTIFICATES

MCP Advanced Topics (Anthropic) | [Certificate](#) || FastAPI for Machine Learning (Campus X) | [Certificate](#)

AI Primer Certification (Infosys) | [Certificate](#) || Principles of Generative AI Certification (Infosys) | [Certificate](#)